Data Analysis Project
P1: Schema Integration
(Group 05)

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Disclaimer: All images used in this document can be found on our repository.

1 Single Source ER Diagrams

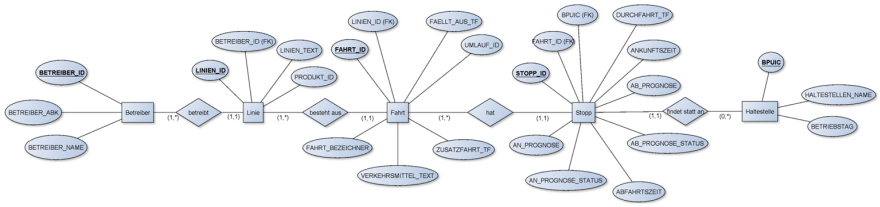


Fig. 1: Entity relationship diagram of the SBB rail data

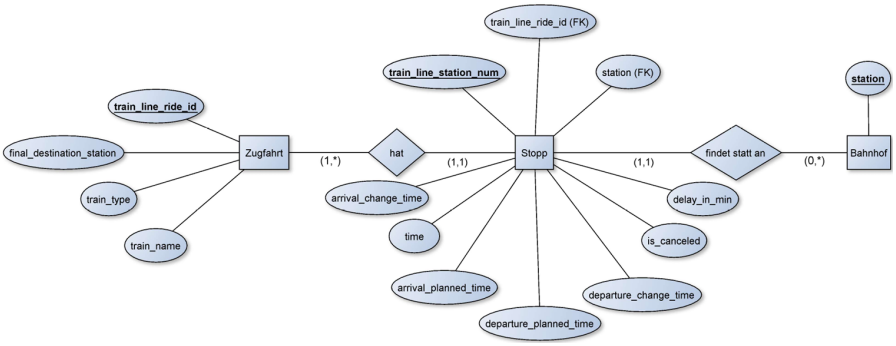


Fig. 2: Entity relationship diagram of the DB rail data

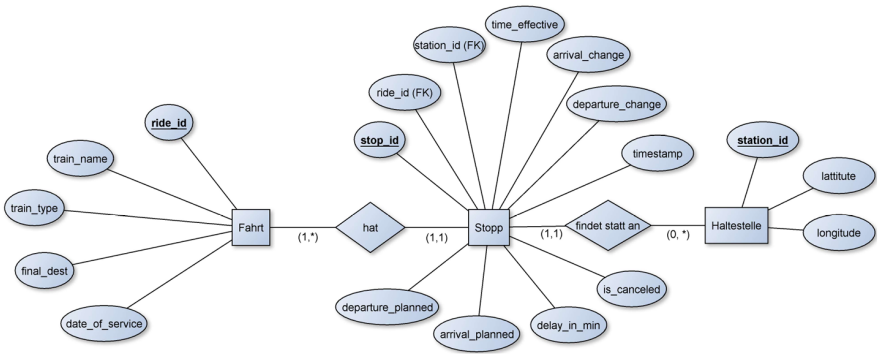


Fig. 3: Entity relationship diagram of the HSP rail data

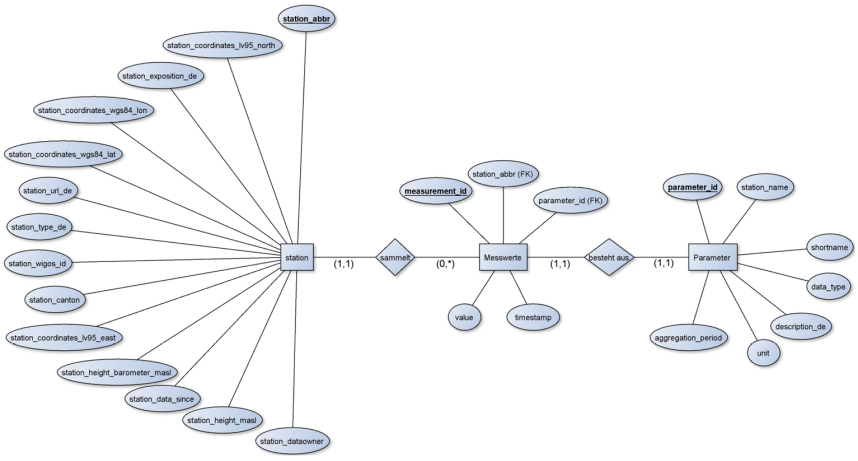


Fig. 4: Entity relationship diagram of the Swiss weather data

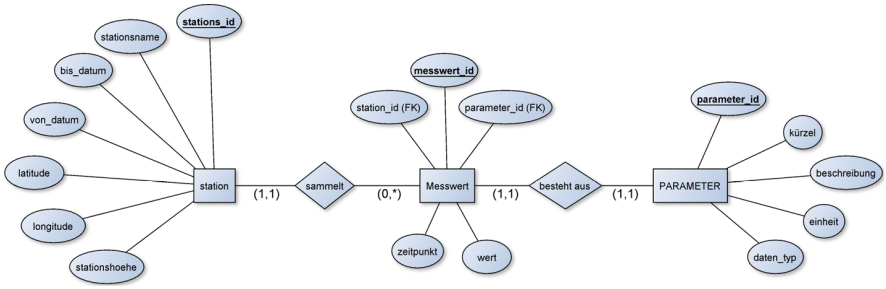


Fig. 5: Entity relationship diagram of the German weather data

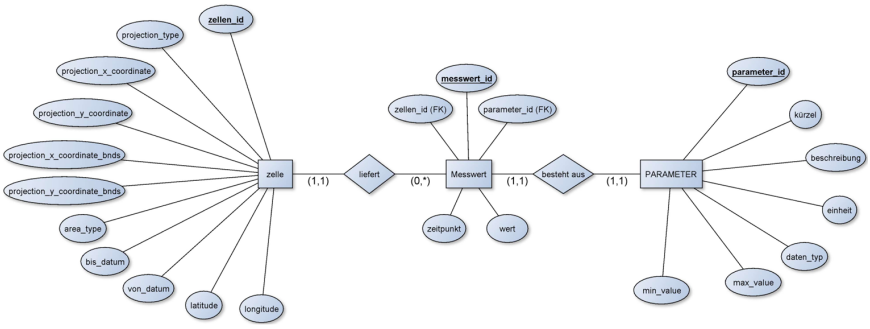


Fig. 6: Entity relationship diagram of the British weather data

2 Integrated ER Diagram

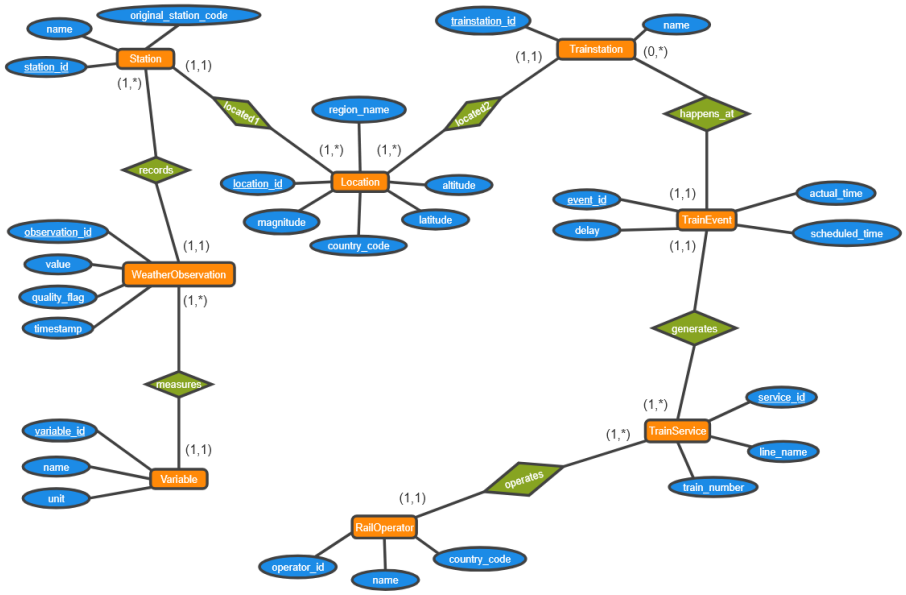


Fig. 7: Entity relationship diagram of the integrated schema

Relational Schema

Stations (station_id, location_id, name, original_station_code)
Variables (variable_id, observation_id, name, unit)
WeatherObservations (observation_id, station_id, value, quality_flag, timestamp)
Locations (location_id, region_name, country_code, altitude, latitude, longitude)
TrainStations (trainstation_id, location_id, name)
TrainEvents (event_id, trainstation_id, service_id, delay, actual_time, scheduled_time)
TrainServices (service_id, line_name, train_number)
RailOperators (operator_id, service_id, name, coutry_code)

3 Normalized Relational Schema

The way we created our integrated ER diagram leads to the resulting relational schema to be already sufficiently normalized. This is because the onlything that derives anything is the artificial primary keys. This directly leads the relational schema to be in BCNF.

As we have not made any changes to the relational schema, the normalized ER diagram is still the one inFigure 7.